Communication-Efficient Federated Bayesian Learning via Client Selection

Jiarong Yang*, Yuan Liu*, and Rahif Kassab†

*Electronic and Information Engineering, South China University of Technology, Guangzhou, China
†Department of Engineering, King's College London, UK
Emails: eejiarong@mail.scut.edu.cn, eeyliu@scut.edu.cn, rahif.kassab@kcl.ac.uk

2022 IEEE Globecom Workshops (GC Wkshps) | 978-1-6654-5975-4/22/$31.00 ©2022 IEEE | DOI: 10.1109/GCWkshps56602.2022.10008503

***Abstract*—Distributed Stein Variational Gradient Descent (DSVGD) is a non-parametric distributed learning framework for federated Bayesian learning, where multiple clients jointly train a machine learning model by communicating a number of non-random and interacting particles with the server. Since communication resources are limited, selecting the clients with most informative local learning updates can improve the model convergence and communication efficiency. In this paper, we propose a client selection scheme for DSVGD based on Hilbert Inner Product (HIP). We derive the upper bound on the decrease of the global free energy per iteration, which is then minimized to speed up the model convergence. We evaluate and compare our scheme with conventional schemes in terms of model accuracy, convergence speed, and stability using various learning tasks and datasets.**

***Index Terms*—Federated Bayesian learning, scheduling, client selection, variational inference.**

I. Introduction

With the increasing awareness for data security and privacy, uploading local data to a remote location is no longer a viable solution [1]–[3]. To cope with this challenge, federated learning [4]–[6] was proposed as a distributed learning framework that performs model training at the client side using the local dataset and then shares the local model with the server, where various models are aggregated. However, most federated learning algorithms are based on frequentist learning which is known to be unable to quantify epistemic uncertainty, yielding overconfident decisions [7], [8]. In contrast, Bayesian learning is more suitable in such scenarios as it provides a more accurate estimate of uncertainty by optimizing over the space of distributions of model parameters instead of a single vector of model parameters [9], [10].

Existing federated Bayesian learning techniques can be classified to two families: Monte Carlo (MC) sampling or Variational Inference (VI). MC-based federated Bayesian learning [11], [12] such as Distributed Stochastic Gradient Langevin Dynamics (DSGLD), maintains a number of Markov chains that are updated via local Stochastic Gradient Descent (SGD) and injects noise into the parameter updates [11], [13]. Unfortunately, the model convergence speed is low. VI-based federated Bayesian learning [14]–[17] such as Partitioned Variational Inference (PVI) optimizes over parametric posteriors from tractable families (e.g., the exponential family) using natural gradient descent [16]. However, limiting the optimization to tractable families incurs bias. Accordingly, Distributed Stein Variational Gradient Descent (DSVGD) has been proposed as a method to eliminate such bias by encoding the variational posterior with deterministic particles that are updated by each selected client using Stein Variational Gradient Descent (SVGD) [18]. When the number of particles is large enough, any posterior could be approximated. By tuning the number of particles, DSVGD can trade off bias, convergence speed and per-iteration complexity [17]. More details on SVGD and DSVGD are given in Section II.

One of the biggest challenges in federated learning is decreasing the communication load. Client selection exploits the difference in the clients' model updates and selects the clients with the most informative updates in order to increase the convergence speed while decreasing the total communication load [19]–[22]. However, most of the existing schemes of client selection in the literature are applicable to frequentist federated learning rather than federated Bayesian learning.

In this paper, we explore the benefits of client selection for federated Bayesian learning and propose a client selection scheme for DSVGD based on the Hilbert Inner Product (HIP) between the SVGD update functions for the global averaged likelihood and local likelihood. We derive the closed form of HIP using the properties of the Reproducing Kernel Hilbert Space (RKHS). The larger the HIP the higher the probability of a client being selected per iteration.

II. System Model

A. Federated Bayesian Learning

As shown in Fig. 1, we consider a system setup for federated Bayesian learning [23], where a central server maintains the variational global posterior distribution $q(\boldsymbol{\theta})$ where $\boldsymbol{\theta}$ is the model parameters vector, for e.g., weights of a neural network. Furthermore, K clients, based on their local dataset D_k, exchange their own view of the variational posterior $q(\boldsymbol{\theta})$ with the server. Following the Bayesian learning framework [24],

The work was supported in part by the Natural Science Foundation of China under Grants 61971196 and U2001210.

Corresponding author: Yuan Liu.

Authorized licensed use limited to: UNIVERSIDADE FEDERAL DE MINAS GERAIS. Downloaded on December 06,2023 at 19:41:17 UTC from IEEE Xplore. Restrictions apply.

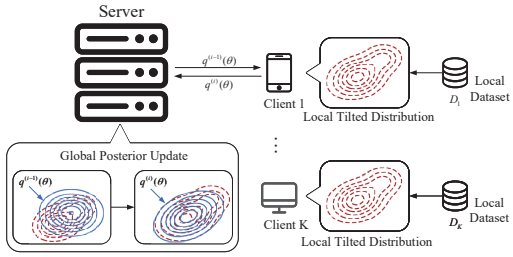


Figure 1: Federated Bayesian learning system.

the goal is to minimize the global free energy over the space of distributions as:

$$\min_{q(\boldsymbol{\theta})} \Bigg\{ F(q(\boldsymbol{\theta})) = \sum_{k=1}^{K} \mathbb{E}_{\boldsymbol{\theta}\sim q(\boldsymbol{\theta})}[L_k(\boldsymbol{\theta})] + \alpha \mathrm{KL}(q(\boldsymbol{\theta}) \| p_0(\boldsymbol{\theta})) \Bigg\}. \quad (1)$$

The global free energy reflects a trade-off between the sum loss function over all clients data and the complexity defined by the Kullback-Leibler (KL) divergence between $q(\boldsymbol{\theta})$ and known prior distribution $p_0(\boldsymbol{\theta})$. The trade-off the two terms is dictated by the temperature level α. Using simple algebraic manipulations, the global free energy in (1) can re-written as

$$\min_{q(\boldsymbol{\theta})} \left\{ F(q(\boldsymbol{\theta})) = \alpha \mathrm{KL}(q(\boldsymbol{\theta}) \| \tilde{q}_{opt}(\boldsymbol{\theta})) \right\}, \quad (2)$$

where $\tilde{q}_{opt}(\boldsymbol{\theta})$ is the unnormalized optimal global posterior define as

$$\tilde{q}_{opt}(\boldsymbol{\theta}) = p_0(\boldsymbol{\theta}) \prod_{k=1}^{K} p_k(\boldsymbol{\theta}), \quad (3)$$

where $p_k(\boldsymbol{\theta})$ is the local likelihood at client k defined as $p_k(\boldsymbol{\theta}) = \prod_{y \in D_k} p(y|\boldsymbol{\theta})$ and $p(y|\boldsymbol{\theta})$ is the likelihood of data point y using model parameters $\boldsymbol{\theta}$. (2) shows visually the optimization target of the VI. Specifically, Bayesian inference aims to obtain the posterior distribution, which is generally referred to as the target distribution, over the model parameters. VI uses KL divergence to measure the similarity between two distributions and frames the Bayesian inference problem into a deterministic optimization that approximates the target distribution with a simpler distribution by minimizing their KL divergence. Therefore, global free energy refers to the KL divergence to be optimized in the VI, which is the KL divergence between the variational global posterior $q(\theta)$ and the optimal global posterior $\tilde{q}_{opt}(\theta)$, as shown in (2).

Following the federated Bayesian learning framework [17], we restrict $q(\boldsymbol{\theta})$ to the family of distributions having the following form: $q(\boldsymbol{\theta}) = p_0(\boldsymbol{\theta}) \prod_{k=1}^{K} t_k(\boldsymbol{\theta})$. The global free energy minimization is then conducted in a distributed manner minimizing the local free energy at each selected client as

$$\min_{q(\boldsymbol{\theta})} \left\{ F_k(q(\boldsymbol{\theta})) = \alpha \mathrm{KL}(q(\boldsymbol{\theta}) \| \tilde{p}_k(\boldsymbol{\theta})) \right\}, \quad (4)$$

where $\tilde{p}_k(\boldsymbol{\theta}) = \frac{q(\boldsymbol{\theta})}{t_k(\boldsymbol{\theta})} p_k(\boldsymbol{\theta})$ is the local tilted distribution and $t_k(\boldsymbol{\theta})$ denotes the approximate scaled local likelihood. It is shown in [16] that minimizing the local free energies (4) across all clients leads to the minimum of the global free energy in (2) by suitably combining the local energies minimums. Once the local free energy minimization problem in (4) is solved by the selected client k at iteration i, a new global variational posterior $q^{(i)}(\boldsymbol{\theta})$ is obtained and the local scaled likelihood is updated as

$$t_k^{(i)}(\boldsymbol{\theta}) = \frac{q^{(i)}(\boldsymbol{\theta})}{q^{(i-1)}(\boldsymbol{\theta})} t_k^{(i-1)}(\boldsymbol{\theta}) \quad (5)$$

for each selected client k, while $t_{k'}^{(i)}(\boldsymbol{\theta}) = t_{k'}^{(i-1)}(\boldsymbol{\theta})$ for clients k' that are not selected.

B. Distributed Stein Variational Gradient Descent

SVGD [18] is a non-parametric variational inference algorithm that can be used to solve problems of a similar form to (4). SVGD encodes the variational distribution $q(\boldsymbol{\theta})$ using a set of particles $\{\boldsymbol{\theta}_n\}_{n=1}^N$ that are updated iteratively to match the (unnormalized) target distribution $\tilde{p}_k(\boldsymbol{\theta})$ as

$$\boldsymbol{\theta}_n^{[l]} \leftarrow \boldsymbol{\theta}_n^{[l-1]} + \epsilon \boldsymbol{\phi}(\boldsymbol{\theta}_n^{[l-1]}), \quad (6)$$

where ϵ is the learning rate and $\boldsymbol{\phi}(\cdot)$ is the SVGD update function for target distribution $\tilde{p}_k(\boldsymbol{\theta})$ defined as

$$\boldsymbol{\phi}(\cdot) = \mathbb{E}_{\boldsymbol{\theta}\sim q}[\nabla \log \tilde{p}_k(\boldsymbol{\theta}) \mathrm{k}(\boldsymbol{\theta}, \cdot) + \nabla \mathrm{k}(\boldsymbol{\theta}, \cdot)] \quad (7)$$

where $\boldsymbol{\phi}(\cdot)$ is restricted to zero-centered balls of a RKHS $\mathcal{H}^d$ defined by a kernel $\mathrm{k}(\cdot, \cdot)$, where the norm of the ball is taken to be $\|\boldsymbol{\phi}\|_{\mathcal{H}^d}^2$.

DSVGD inherits the performance of SVGD in terms of bias, convergence speed, and per-iteration complexity by controlling the number of particles [17]. At each communication round, also referred to as global iteration $i = 1, 2, \ldots, I$, DSVGD maintains a set of global particles $\{\boldsymbol{\theta}_n^{(i-1)}\}_{n=1}^N$ at the server side over which the approximate posterior $q^{(i-1)}(\boldsymbol{\theta})$ of the current model parameters is obtained using a Kernel Density Estimator (KDE) as $q^{(i-1)}(\boldsymbol{\theta}) = \sum_{n=1}^N K(\boldsymbol{\theta}, \boldsymbol{\theta}_n^{(i-1)})$. Furthermore, DSVGD maintains a set of local particles $\{\boldsymbol{\theta}_{n,k}^{(i-1)}\}_{n=1}^N$ at the client side and uses KDE to represent the local approximate likelihood $t_k^{(i-1)}(\boldsymbol{\theta})$. Both the global and local particles are updated using SVGD as follows: first, the global particles $\{\boldsymbol{\theta}_n^{(i-1)}\}_{n=1}^N$ are downloaded by the client k and updated by running L local SVGD iterations to approximate the local tilted distribution $\tilde{p}_k^{(i)}(\boldsymbol{\theta})$ as

$$\boldsymbol{\theta}_n^{[l]} \leftarrow \boldsymbol{\theta}_n^{[l-1]} + \frac{\epsilon}{N} \sum_{j=1}^{N} [\mathrm{k}(\boldsymbol{\theta}_j^{[l-1]}, \boldsymbol{\theta}_n^{[l-1]}) \nabla_{\boldsymbol{\theta}_j} \log \tilde{p}_k^{(i)}(\boldsymbol{\theta}_j^{[l-1]}) + \nabla_{\boldsymbol{\theta}_j} \mathrm{k}(\boldsymbol{\theta}_j^{[l-1]}, \boldsymbol{\theta}_n^{[l-1]})], \quad (8)$$

for $l = 1, 2, \ldots, L$. Note that we denote the SVGD algorithm iterations by $[l]$ and the communication rounds by (i). Simi-

Authorized licensed use limited to: UNIVERSIDADE FEDERAL DE MINAS GERAIS. Downloaded on December 06,2023 at 19:41:17 UTC from IEEE Xplore. Restrictions apply.

larly, the local particles are updated by running L' local SVGD iterations to approximate the distribution $t_k^{(i)}(\boldsymbol{\theta})$ as

$$\boldsymbol{\theta}_{k,n}^{[l']} \leftarrow \boldsymbol{\theta}_{k,n}^{[l'-1]} + \frac{\epsilon'}{N}\sum_{j=1}^{N}[\mathrm{k}(\boldsymbol{\theta}_{k,j}^{[l'-1]}, \boldsymbol{\theta}_{k,n}^{[l'-1]})\nabla_{\boldsymbol{\theta}_j}\log t_k^{(i)}(\boldsymbol{\theta}) + \nabla_{\boldsymbol{\theta}_j}\mathrm{k}(\boldsymbol{\theta}_{k,j}^{[l'-1]}, \boldsymbol{\theta}_{k,n}^{[l'-1]})], \quad (9)$$

where $t_k^{(i)}(\boldsymbol{\theta})$ is computed using (5). To summarize, the process of DSVGD is as follows:

- **Step 1:** The server serially selects a client and sends the current global particles $\{\boldsymbol{\theta}_n^{(i-1)}\}_{n=1}^N$ to the selected client.
- **Step 2:** The selected client minimizes the local free energy in (4) by running SVGD algorithm (see (8)) to obtain the current global particles $\{\boldsymbol{\theta}_n^{(i)}\}_{n=1}^N$.
- **Step 3:** The updated global particles $\{\boldsymbol{\theta}_n^{(i)}\}_{n=1}^N$ are uploaded to the server and the local particles $\{\boldsymbol{\theta}_{n,k}^{(i-1)}\}_{n=1}^N$ are updated by running SVGD algorithm (see (9)) to obtain the current local particles $\{\boldsymbol{\theta}_{n,k}^{(i)}\}_{n=1}^N$ at the client side.
- **Step 4:** The server updates the current global particles by setting $\{\boldsymbol{\theta}_n^{(i)}\}_{n=1}^N \longleftarrow \{\boldsymbol{\theta}_n^{(i-1)}\}_{n=1}^N$ and continues to serially select the next client.

In this paper, we consider the selection problem for DSVGD which aims to select the client that contributes the most to the decrease of the global free energy in each global iteration and thus improving convergence.

III. Client Selection Scheme for Federated Bayesian Learning

A. Generic Client Selection Framework

We now introduce the generic client selection framework for federated Bayesian learning where the server estimates the selection distribution $\boldsymbol{P}^{(i)} = (P_1^{(i)}, P_2^{(i)}, ..., P_k^{(i)})$ which represents the probability of selecting client k. At each global iteration i, a client $k \in \{1, 2, ..., K\}$ is selected based on the distribution $\boldsymbol{P}^{(i)}$ to update the global posterior. The step of this framework can be summarized as follows:

- **Step 1** (Global Posterior Broadcast): The server broadcasts the current global posterior to all clients.
- **Step 2** (Importance Indicator Report): Each client k computes an indicator and reports the result to the server. This step is going to be detailed later.
- **Step 3** (Selection Distribution Calculation): Based on the received reports, the server computes the selection distribution $\boldsymbol{P}^{(i)}$ of the clients.
- **Step 4** (Client Selection): Based on the selection distribution, the server selects a client, and then solicits updates from the selected client.
- **Step 5** (Global Posterior Update): The selected client minimizes local free energy in (8) to obtain the new global posterior.
- **Step 6** (Global Posterior Upload and Local Approximate Likelihood Update): The selected client uploads the global posterior obtained in the previous step to the server and updates the local approximate likelihood using (9).

Note that the client selection requires the addition of three steps: the importance indicator report (Step 2), selection distribution calculation (Step 3) and client selection (Step 4). Also, the global posterior needs to be broadcast to all clients at the beginning of each communication iteration, instead of being sent serially to each client.

B. HIP-based scheme

In this subsection, we propose the HIP-based scheme with two principles. First, the HIP refers to the inner product of the local SVGD update function and the SVGD update function for the global averaged likelihood $p' = \left(\prod_{m=1}^K p_m\right)^{\frac{1}{K}}$ in the Hilbert space defined by a kernel function. Note that p' contains the likelihood information of all clients. From the nature of the inner product, the larger the inner product is, the more similar the update direction of the local model is to that of the global model. Therefore, prioritizing the selection of a client with larger HIP could improve the performance. Second, the upper bound of the decrease of the global free energy per iteration is related to the HIP of clients, and the larger the HIP the smaller the upper bound, indicating that the client is able to make the decrease of the global free energy at a faster rate, thus improving the performance of the system. Next, we detail the two points sequentially.

1) Nature of HIP: We start by recalling the update function for particles in SVGD in (6). The update function is restricted to a zero-centered ball of a RKHS $\mathcal{H}^d$ and is optimized to maximize the negative gradient of the KL divergence, which can be written in the form of the HIP as $\langle \boldsymbol{\phi}, \boldsymbol{\phi}^* \rangle_{\mathcal{H}^d}$, where $\boldsymbol{\phi}^*(\cdot) = \mathbb{E}_{\boldsymbol{\theta}\sim q}[\nabla \log \tilde{p}_k(\boldsymbol{\theta})\mathrm{k}(\boldsymbol{\theta}, \cdot) + \nabla \mathrm{k}(\boldsymbol{\theta}, \cdot)]$ is referred to as the target update function for normalized distribution $p_k(\boldsymbol{\theta}) \propto \tilde{p}_k(\boldsymbol{\theta})$ [18]. When the update function $\boldsymbol{\phi}(\boldsymbol{\theta})$ is equal in size and direction to the target update function $\boldsymbol{\phi}^*(\boldsymbol{\theta})$, the HIP can achieve a maximum value, i.e., when the norm of the ball is taken to be $\|\boldsymbol{\phi}^*\|_{\mathcal{H}^d}^2$. Similarly, when we choose a different update function, the difference in both the size and direction between it and the target update function in Hilbert space will cause the HIP to change, which in turn affects the speed at which the particles are updated towards the target distribution. Therefore, we can use the HIP between the update function $\boldsymbol{\phi}(\boldsymbol{\theta})$ and the target update function $\boldsymbol{\phi}(\boldsymbol{\theta}^*)$ to measure the contribution of the local particles update at each selected client. A larger HIP indicates that the update function can make the variational distribution approach the target distribution at a faster speed. The estimation of the HIP between two SVGD update functions is obtained using the following theorem.

Theorem 1. *Given i.i.d. samples* $\{\boldsymbol{\theta}_n\}_{n=1}^N$ *drawn from the distribution* $q(\boldsymbol{\theta})$ *and the score function* S_{p1} *and* S_{p2}*. We can estimate the HIP between the SVGD update function* $\boldsymbol{\phi}_1(\cdot) = \mathbb{E}_{\boldsymbol{\theta}\sim q}[\nabla \log p_1(\boldsymbol{\theta})\mathrm{k}(\boldsymbol{\theta}, \cdot) + \nabla \mathrm{k}(\boldsymbol{\theta}, \cdot)]$ *for* p_1 *and the SVGD update function* $\boldsymbol{\phi}_2(\cdot) = \mathbb{E}_{\boldsymbol{\theta}\sim q}[\nabla \log p_2(\boldsymbol{\theta})\mathrm{k}(\boldsymbol{\theta}, \cdot) + \nabla \mathrm{k}(\boldsymbol{\theta}, \cdot)]$

Authorized licensed use limited to: UNIVERSIDADE FEDERAL DE MINAS GERAIS. Downloaded on December 06,2023 at 19:41:17 UTC from IEEE Xplore. Restrictions apply.

for p_2 by using the properties of the RKHS. Define

$$h_{p_1,p_2}(\boldsymbol{\theta}_i,\boldsymbol{\theta}_j) = S_{p_1}(\boldsymbol{\theta}_i)^T \mathrm{k}(\boldsymbol{\theta}_i,\boldsymbol{\theta}_j) S_{p_2}(\boldsymbol{\theta}_j) + S_{p_1}(\boldsymbol{\theta}_i)^T \nabla_{\boldsymbol{\theta}_j} \mathrm{k}(\boldsymbol{\theta}_i,\boldsymbol{\theta}_j) + \nabla_{\boldsymbol{\theta}_i} \mathrm{k}(\boldsymbol{\theta}_i,\boldsymbol{\theta}_j)^T S_{p_2}(\boldsymbol{\theta}_j) + \mathrm{trace}(\nabla_{\boldsymbol{\theta}_i,\boldsymbol{\theta}_j} \mathrm{k}(\boldsymbol{\theta}_i,\boldsymbol{\theta}_j)),$$

then the HIP can be estimated as

$$\langle \boldsymbol{\phi}_1, \boldsymbol{\phi}_2 \rangle_{\mathcal{H}^d} = \frac{1}{N^2} \sum_{i,j=1}^{N} [h_{p_1,p_2}(\boldsymbol{\theta}_i, \boldsymbol{\theta}_j)]. \tag{10}$$

Theorem 1 reveals that we can estimate the HIP between the SVGD update function $\boldsymbol{\phi}_1$ for p_1 and the target update function $\boldsymbol{\phi}_2$ for target distribution p_2 using (10), and thus use the HIP estimate to measure the contribution of the SVGD update function for p_1 to the update of particles towards the target distribution p_2. Therefore, we define the global averaged likelihood $p' = \left(\prod_{m=1}^{K} p_m\right)^{\frac{1}{K}}$ as the target distribution with the SVGD update function as $\boldsymbol{\phi}'(\cdot) = \mathbb{E}_{\boldsymbol{\theta}\sim q}[\nabla \log p'(\boldsymbol{\theta}) \mathrm{k}(\boldsymbol{\theta}, \cdot) + \nabla \mathrm{k}(\boldsymbol{\theta}, \cdot)]$. Accordingly, the contribution of client k to the update of particles towards the global averaged likelihood can be measured by the HIP as $\langle \boldsymbol{\phi}_k, \boldsymbol{\phi}' \rangle_{\mathcal{H}^d}$.

2) Convergence Analysis: Federated Bayesian learning converges as the global free energy decreases. According to [17], we have the upper bound of the decrease of the global free energy per local iteration as

$$\left(F(q^{[l+1]}(\boldsymbol{\theta})) - F(q^{[l]}(\boldsymbol{\theta}))\right)/\alpha \leq -\epsilon S(q^{[l]}, \tilde{p}_k) + 2(K-1) l_{max}^{(i)} \sqrt{2\mathrm{KL}(q^{[l+1]} \| q^{[l]})}, \tag{11}$$

where $l_{max}^{(i)} = \sup_{\boldsymbol{\theta}} \max_{m \neq k} \left| \log(t_m^{(i-1)}(\boldsymbol{\theta})) \cdot p_m(\boldsymbol{\theta}) \right|$. The first term on the right hand side is expressed by the kernel Stein discrepancy [25] between the variational global posterior and the local tilted distribution of selected client, while the second term corresponds to the upper bound of the impact of un-selected clients on the global free energy. Note that the second term is intractable, so speeding up the convergence by minimizing the upper bound is hard to achieve. Therefore, we rewrite the upper bound by introducing the HIP, as shown in the following theorem.

Theorem 2. *Given $\langle \boldsymbol{\phi}_k, \boldsymbol{\phi}' \rangle_{\mathcal{H}^d}$ representing the HIP between the SVGD update functions for the likelihood p_k of client k and the global averaged likelihood p', the upper bound of the decrease of the global free energy per iteration in* (11) *can be re-written in terms of the HIP as*

$$\left(F(q^{[l+1]}(\boldsymbol{\theta})) - F(q^{[l]}(\boldsymbol{\theta}))\right)/\alpha \leq -\epsilon K \cdot \langle \boldsymbol{\phi}_k, \boldsymbol{\phi}' \rangle_{\mathcal{H}^d} + \sum_{m=1}^{K} \mathrm{KL}(q^{[l]} \| t_m^{(i-1)}). \tag{12}$$

Here in Theorem 2, and in contrast with (11), only the first term in the upper bound is related to the selected client, and thus, the second term can be ignored. Theorem 2 reveals that the upper bound is smaller when the HIP term $\langle \boldsymbol{\phi}_k, \boldsymbol{\phi}' \rangle_{\mathcal{H}^d}$ is larger, indicating that the client has a greater contribution to the global particles update if the client can get the particles to update towards the averaged likelihood p', which contains information about all clients, at a faster speed. Therefore, we consider the HIP term $\langle \boldsymbol{\phi}_k, \boldsymbol{\phi}' \rangle_{\mathcal{H}^d}$ as a measure of the importance of the clients' data and assign higher selection probabilities to the clients with larger HIP. This is formalized in the definition below. The HIP-based selection distribution is defined as:

$$P_{\mathrm{HIP},k} = \frac{\langle \boldsymbol{\phi}_k, \boldsymbol{\phi}' \rangle_{\mathcal{H}^d}}{\sum_{m=1}^{K} \langle \boldsymbol{\phi}_m, \boldsymbol{\phi}' \rangle_{\mathcal{H}^d}}. \tag{13}$$

We use the HIP between the update function for local likelihood p_k and the update function for the averaged likelihood p' as an indicator for judging the importance of a client's data. Intuitively, using (13), the server can prioritize the client who updates particles towards the averaged likelihood at a faster speed, maximizing the decrease in global free energy per iteration. Finally, we present the HIP-based scheme in Algorithm 1, where lines 4-6 corresponds the extra steps with respect to DSVGD and are added for client selection.

Algorithm 1 HIP-based client selection scheme for DSVGD

Input: Prior $p_0(\boldsymbol{\theta})$, local likelihood $\{p_k\}_{k=1}^K$, temperature $\alpha > 0$, kernels $\mathrm{K}(\cdot,\cdot)$ and $\mathrm{k}(\cdot,\cdot)$.

Output: Global approximate posterior $q(\boldsymbol{\theta}) = N^{-1} \sum_{n=1}^{N} \mathrm{K}(\boldsymbol{\theta}, \boldsymbol{\theta}_n)$.

1: **initialize** $q^{(0)}(\boldsymbol{\theta}) = p_0(\boldsymbol{\theta})$; $\{\boldsymbol{\theta}_n^{(0)}\}_{n=1}^N \overset{\text{i.i.d.}}{\sim} p_0(\boldsymbol{\theta})$; $\{\boldsymbol{\phi}_n^{(0)} = \boldsymbol{\theta}_{k,n}^{(0)} = \boldsymbol{\theta}_n^{(0)}\}_{n=1}^N$ and $t_k^{(0)}(\boldsymbol{\theta}) = 1$ for $k = 1, ..., K$.
2: **for** $i = 1, \cdots, I$ **do**
3: Server sends $\{\boldsymbol{\theta}_n^{(i-1)}\}_{n=1}^N$ to all clients;
4: Each client k computes its own score function gradient $\{\nabla_{\boldsymbol{\theta}_n} \log p_k\}_{n=1}^N$;
5: Each client k sends $\{\nabla_{\boldsymbol{\theta}_n} \log p_k\}_{n=1}^N$ back to server;
6: Server computes the client selection probability $P_{\mathrm{HIP},k}^{(i)}$ using (1), (13) and $\{\nabla_{\boldsymbol{\theta}_n} \log p_k\}_{n=1}^N, k = 1, ..., K$;
7: Server randomly selects a client k according to $P_{\mathrm{HIP},k}^{(i)}, k = 1, ..., K$;
8: Client k obtains updated global particles $\{\boldsymbol{\theta}_n^{(i)}\}_{n=1}^N$ according to (8);
9: Client k sends the updated global particles $\{\boldsymbol{\theta}_n^{(i)}\}_{n=1}^N$ to server and obtains updated local particles $\{\boldsymbol{\theta}_{k,n}^{(i)}\}_{n=1}^N$ according to (9).
10: **end for**
11: **return** $q(\boldsymbol{\theta}) = N^{-1} \sum_{n=1}^{N} \mathrm{K}(\boldsymbol{\theta}, \boldsymbol{\theta}_n^{(I)})$.

IV. Experimental Results

A. Experiment Settings

All of our experiments with SVGD and DSVGD were performed with the same settings as in [17] and [18]. We use the Radial Basis Function (RBF) kernel $\mathrm{k}(x, x_0) = exp(-\|x - x_0\|_2^2/h)$ and take the bandwidth to be $h = \mathrm{med}^2/\log n$ where med is the median of the pairwise distance between

Authorized licensed use limited to: UNIVERSIDADE FEDERAL DE MINAS GERAIS. Downloaded on December 06,2023 at 19:41:17 UTC from IEEE Xplore. Restrictions apply.

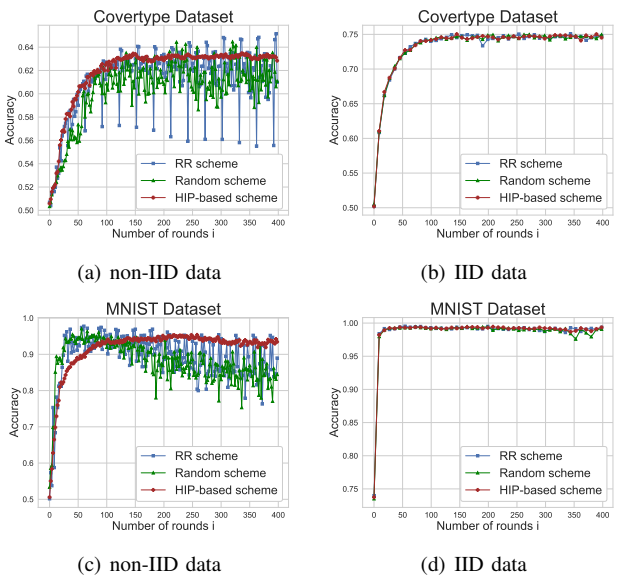
the particles in the current iterate, so that the bandwidth h changes according to the set of particles adaptively across the iterations. As for the KSD estimator, we use the Gaussian kernel $\mathrm{K}(\cdot,\cdot)$ where the bandwidth is set to 0.55. We use AdaGrad for step size and initialize the particles using the prior distribution unless otherwise specified. Finally, we fix the temperature parameter $\alpha = 1$ in (1) throughout the iterations.

For exposition, we consider the learning task of training classifiers. Two prevalent learning models of Bayesian logistic regression (BLR) [26] and Bayesian Neural Networks (BNN) [27] are employed for implementation. First, We consider BLR model for binary classification using the same setting as in [26]. The model parameters is define as $\boldsymbol{\theta} = [\mathbf{w}, \log(\xi)]$ where parameter ξ is a precision and the regression weights $\mathbf{w}$ is assigned with a Gaussian prior $p_0(\mathbf{w}|\xi) = \mathcal{N}(\mathbf{w}|\mathbf{0}, \xi^{-1}\mathbf{I}_d)$ and $p_0(\xi) = \mathrm{Gamma}(\xi|1, 0.01)$. The local likelihood at each client k is given as $p_k = \prod_{x_i,y_i \in D_k} p(y_i|x_i, \mathbf{w}, \xi)$, where D_k is the dataset at client k with covariates $x_k \in \mathbb{R}^d$ and label $y_k \in \{-1, 1\}$. Similarly, we consider multi-label classification with BNN model using the same setting as in [17]. The model parameters $\boldsymbol{\theta} = [\mathbf{W}, \log(\gamma)]$ include a precision parameter γ along with the weights of neural network $\mathbf{W}$ assigned with a Gaussian prior $p_0(\mathbf{W}|\lambda) = \mathcal{N}(\mathbf{W}|\mathbf{0}, \lambda^{-1}\mathbf{I}_d)$ with a fixed precision $\lambda = e$. The data of client k is given as $D_k = \{x_i, y_i\}_{i=1}^K$ and y_n is obtained as $y_n = f(x_n; \mathbf{W}) + n$ where the neural network output $f(\cdot; \mathbf{W})$ is corrupted by additive noise variables n, where $n \sim \mathcal{N}(0, \gamma^{-1})$. The number of hidden layers containing 100 hidden units is set to one and RELU is set as the activation function. The local likelihood at each client k is given as $p_k = \prod_{x_i,y_i \in D_k} p(y_i|x_i, \mathbf{W}, \gamma)$.

To demonstrate the effectiveness of the proposed selection schemes, we implement two baseline schemes in the following experiments, the round robin (RR) selection scheme and the random selection scheme. Specifically, the RR selection scheme is used by the authors of [17] in DSVGD, which selects each client in a certain order, and the random selection scheme is the more common selection scheme in federated learning, where the server randomly selects a client at each iteration. In addition, since the communication load caused by HIP-based selection scheme is similar to that of the parallel DSVGD algorithm, we also conduct experiment to compare the model performance between the two schemes.

B. Bayesian Logistic Regression

We now consider BLR for binary classification using the Covertype dataset and the MNIST dataset. To better simulate the data distribution, we consider the non-IID data partitioning way as follows. For the Covertype dataset, we assign each client 90% of one type and 10% of the other type. For the MNIST dataset, we select images labelled as "1" and "2" for model training, where each client is assigned 97% of one type and 3% of the other type. From Fig. 2, we observe the better performance of HIP-based scheme compared to that of two baseline schemes using the BLR model. Specifically, with HIP-based scheme in the Covertype dataset, the model accuracy curve is always stable compared to the two baseline

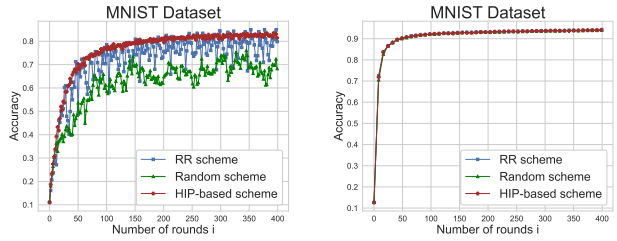


(a) non-IID data (b) IID data

(c) non-IID data (d) IID data

Figure 2: Performance comparison using the BLR model.

schemes and the model accuracy is almost always higher than that of two baseline schemes, which especially significant in late iterations. In the MNIST dataset, the accuracy of two baseline schemes is higher than that of the proposed scheme in the early stages of model training, however, this does not represent the true model convergence accuracy because two baseline schemes occur the oscillation or ping-pong effect during the iterations, and the model accuracy is decreasing as the training proceeds. While the HIP-based scheme significantly improves the model performance and the model converges stably with high accuracy at the later stages of training. Note that the proposed scheme performs similarly with the generic schemes in the IID environment, since each client contributes equally to the model updates.



(a) non-IID data (b) IID data

Figure 3: Performance comparison using the BNN model.

C. Bayesian Neural Networks

We now consider multi-label classification with BNN model using MNIST dataset. Here we also consider the non-IID data distribution where each client owns only three types of labelled images. We now consider multi-label classification with BNN model using MNIST dataset. Here we also consider the non-IID data distribution where each client owns only three types of labelled images. We plot the convergence curves of the proposed scheme and the two baseline schemes in Fig. 3. It

Authorized licensed use limited to: UNIVERSIDADE FEDERAL DE MINAS GERAIS. Downloaded on December 06,2023 at 19:41:17 UTC from IEEE Xplore. Restrictions apply.

can be seen that the proposed scheme brings improvements in convergence speed, model accuracy and model convergence stability, demonstrating that the HIP-based scheme significantly improves the model performance in an environment with a pathological non-IID client data distribution. Note that the proposed scheme no longer improves model performance in IID datasets, since each client contributes equally to the model updates.

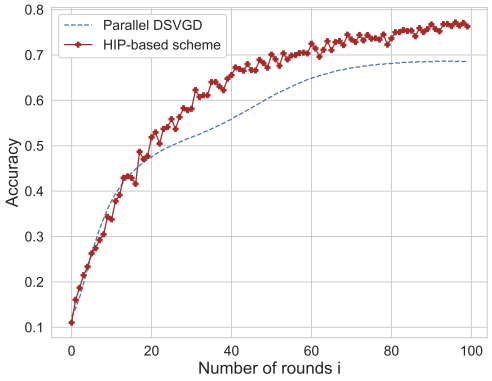


Figure 4: Performance comparison between the HIP-based and the Parallel DSVGD algorithm using the BNN model.

D. Comparison with the Parallel DSVGD Algorithm

Fig. 4 depicts the performance of the HIP-based scheme compared with the Parallel DSVGD algorithm using the BNN model. From this figure, we can observe that the convergence curve of the Parallel DSVGD algorithm is more stable than that of the the HIP-based scheme. The reason is that the parallel DSVGD algorithm can use the likelihood information of all clients, so that the global posterior does not favor a local posterior. However, as the training proceeds, the model accuracy of the HIP-based scheme becomes gradually higher than that of the Parallel DSVGD algorithm. The reason is that in the Parallel DSVGD algorithm, there is an error in the KDE of the local particles uploaded at the client side, which does not accurately represent the information of the local approximate likelihood, thus causing the loss of client data information in this step and making the accuracy of the model degrade.

V. Conclusion

In this paper, we proposed a client selection scheme based on HIP for federated Bayesian learning. Through convergence analysis, we found that the HIP-based scheme improves the model convergence by maximizing the decrease of the global free energy per iteration. Comprehensive experiments using real datasets confirmed the performance gain of the proposed scheme compared with the baseline schemes.

References

[1] Z. Liang, Y. Liu, T.-M. Lok, and K. Huang, "Multi-cell mobile edge computing: Joint service migration and resource allocation," *IEEE Trans. Wireless Commun.*, vol. 20, no. 9, pp. 5898–5912, 2021.

[2] M. Liu and Y. Liu, "Price-based distributed offloading for mobile-edge computing with computation capacity constraints," *IEEE Wireless Commun. Lett.*, vol. 7, no. 3, pp. 420–423, 2017.

[3] Z. Liang, Y. Liu, T.-M. Lok, and K. Huang, "Multiuser computation offloading and downloading for edge computing with virtualization," *IEEE Trans. Wireless Commun.*, vol. 18, no. 9, pp. 4298–4311, 2019.

[4] Q. Yang, Y. Liu, T. Chen, and Y. Tong, "Federated machine learning: Concept and applications," *ACM Trans. Intell. Syst. Technol.*, vol. 10, no. 2, Jan. 2019.

[5] M. Chen, Z. Yang, W. Saad, C. Yin, H. V. Poor, and S. Cui, "A joint learning and communications framework for federated learning over wireless networks," *IEEE Trans. Wireless Commun.*, vol. 20, no. 1, pp. 269–283, 2020.

[6] Z. Yang, M. Chen, W. Saad, C. S. Hong, and M. Shikh-Bahaei, "Energy efficient federated learning over wireless communication networks," *IEEE Trans. Wireless Commun.*, vol. 20, no. 3, pp. 1935–1949, 2020.

[7] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. Intl. Conf. Mach. Learning (ICML)*, (Sydney, Australia), Aug. 2017, pp. 1321–1330.

[8] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Proc. Advances Neural Inf. Process. Syst.*, 2017, pp. 6402–6413.

[9] D. Barber, "Bayesian reasoning and machine learning," 2012.

[10] O. Simeone, *A Brief Introduction to Machine Learning for Engineers*, 2018.

[11] S. Ahn, B. Shahbaba, and M. Welling, "Distributed stochastic gradient mcmc," in *Proc. Int. Conf. Mach. Learn.*, 2014, pp. 1044–1052.

[12] D. Mesquita, P. Blomstedt, and S. Kaski, "Embarrassingly parallel mcmc using deep invertible transformations," in *Proceedings of The 35th Uncertainty in Artificial Intelligence Conference*, 2020, pp. 1244–1252.

[13] M. Welling and Y. W. Teh, "Bayesian learning via stochastic gradient langevin dynamics," in *Proc. 28th Int. Conf. Mach. Learn.*, 2011, pp. 681–688.

[14] E. Angelino, M. J. Johnson, and R. P. Adams, "Patterns of scalable bayesian inference," 2016. [Online]. Available: https://arxiv.org/pdf/1602.05221.pdf

[15] L. Corinzia, A. Beuret, and J. M. Buhmann, "Variational federated multi-task learning," 2019. [Online]. Available: https://arxiv.org/pdf/1906.06268.pdf

[16] T. D. Bui, C. V. Nguyen, S. Swaroop, and R. E. Turner, "Partitioned variational inference: A unified framework encompassing federated and continual learning," 2018. [Online]. Available: https://arxiv.org/pdf/1811.11206.pdf

[17] R. Kassab and O. Simeone, "Federated generalized bayesian learning via distributed stein variational gradient descent," 2020. [Online]. Available: https://arxiv.org/pdf/2009.06419.pdf

[18] Q. Liu and D. Wang, "Stein variational gradient descent: A general purpose bayesian inference algorithm," in *Proc. 30th Int. Conf. Neural Inf. Process. Syst.*, 2016, p. 2378–2386.

[19] P. Zhao and T. Zhang, "Stochastic optimization with importance sampling for regularized loss minimization," in *Proc. 32nd Int. Conf. Mach. Learn.* PMLR, 2015, pp. 1–9.

[20] D. Liu, G. Zhu, J. Zhang, and K. Huang, "Data-importance aware user scheduling for communication-efficient edge machine learning," *IEEE Trans. Cogn. Commun. Netw.*, vol. 7, no. 1, pp. 265–278, Jun. 2021.

[21] J. Ren, Y. He, D. Wen, G. Yu, K. Huang, and D. Guo, "Scheduling for cellular federated edge learning with importance and channel awareness," *IEEE Trans. Wireless Commun.*, vol. 19, no. 11, pp. 7690–7703, 2020.

[22] H. T. Nguyen, V. Sehwag, S. Hosseinalipour, C. G. Brinton, M. Chiang, and H. Vincent Poor, "Fast-convergent federated learning," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 1, pp. 201–218, 2021.

[23] A. Vehtari, A. Gelman, T. Sivula, P. Jylänki, D. Tran, S. Sahai, P. Blomstedt, J. P. Cunningham, D. Schiminovich, and C. P. Robert, "Expectation propagation as a way of life: A framework for bayesian inference on partitioned data," *J. Mach. Learn. Res.*, vol. 21, pp. 1–53, 2020.

[24] J. Knoblauch, J. Jewson, and T. Damoulas, "Generalized variational inference," *stat*, vol. 1050, p. 21, 2019.

[25] Q. Liu, J. Lee, and M. Jordan, "A kernelized stein discrepancy for goodness-of-fit tests," in *Proc. 33rd Int. Conf. Mach. Learn.*, 2016, pp. 276–284.

[26] S. Gershman, M. D. Hoffman, and D. M. Blei, "Nonparametric variational inference." in *Proc. 29th Int. Conf. Mach. Learn.*, 2012.

[27] J. M. Hernández-Lobato and R. Adams, "Probabilistic backpropagation for scalable learning of bayesian neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 1861–1869.

Authorized licensed use limited to: UNIVERSIDADE FEDERAL DE MINAS GERAIS. Downloaded on December 06,2023 at 19:41:17 UTC from IEEE Xplore. Restrictions apply.